LXS Demura IP C Library Guide

LX89507

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2026.04.09

Laboratory
Image IP Team

History

No.	Date	Description	Author
1	2026.02.24	Initial.	LXS
2	2026.03.19	• The Corner-Cut functions have been updated, including subdividing CORNER_CUT_TOP into CORNER_CUT_TOP_H and CORNER_CUT_TOP_V. • Information related to Total_crc.bin has been updated.	LXS
3	2026.04.09	• TOTAL_CRC file generation ON/OFF option has been updated.	LXS

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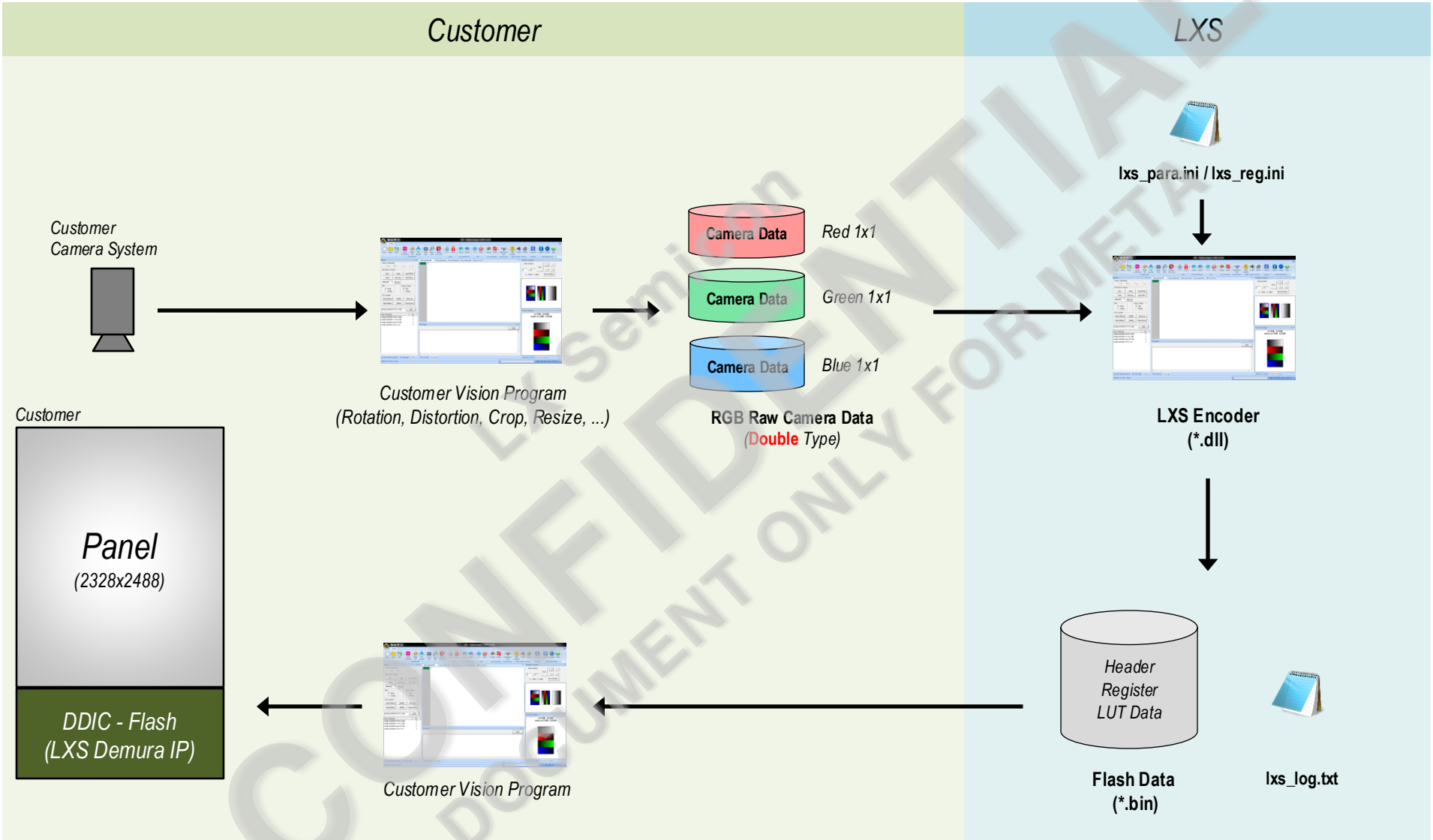
LXS Demura Process Flow



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Demura Library Interface Guide

Program Folder Hierarchy

Location

- *Customer.exe*
- *LX89507_Demura.dll* — *lxs_demura static link library.*
- *lxs_reg.ini* — *Set Registers for the LXS C library.* — *The user can change to tune the values of all registers*
- *lxs_para.ini* — *Set Parameters for the LXS C Library.* — *"Registers" are stored in a binary file(.bin), while "Parameters" are used in a DLL to create a LUT and are not stored in a binary file(.bin).*

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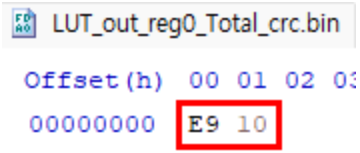
Flash : Binary File Configuration

- **Flash Size**
  - *Header Size* : *4 Byte*
  - *Demura CFG Size* : *256 Byte*
  - *CRC for Header and CFG* : *2 Byte*
  - *LUT data Size* : *Variable*
  - *CRC for LUT* : *2 Byte*

Start Address	Memory Region
23'h00000	*Header (4 Byte)*
23'h00004	*Demura CFG Data (256 Byte)*
23'h00104	*CRC for header & CFG (2 Byte)*
23'h00106	*Demura LUT Data (Variable)*
23'h????? (Flexible)	*CRC for LUT (2 Byte)*

- **CRC (Cyclic Redundancy Check)**
  - *CRC-16 Modbus*
    - *Polynomial : 0xA001*
    - *Initial Value: 0xFFFF*
    - *Reflection: Both the input data and the output CRC value*
    - *Bit Width : 16*
  - *Total_crc.bin*
    - *Size: 2 bytes (big-endian)*
    - *Algorithm: CRC-16 Modbus (polynomial 0xA001, init 0xFFFF)*
    - *Purpose: End-to-end integrity check of the entire demura output data (CFG + LUT)*



Total_crc.bin

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Flash : Binary File Configuration

- **Flash Binary File (.bin)**

Resolution 2328x2488
(Only for LX89507)

Address	Data
23'h00000	*Header (4 Bytes)*
23'h00004	*Demura CFG Data (256 Bytes)*
23'h00104	*CRC for header & CFG (2 Bytes)*
23'h00106	*Dummy*
23'h00108	*Demura LUT Data (73,032 Bytes)*
23'h11E50	*CRC for Demura LUT (2 Bytes)*
23'h11E52	*Dummy*

Demura Mode0
(RGB Mode, Plane 2, Compression)

Address	Data
23'h00000	*Header (4 Bytes)*
23'h00004	*Demura CFG Data (256 Bytes)*
23'h00104	*CRC for header & CFG (2 Bytes)*
23'h00106	*Dummy*
23'h00108	*Demura LUT Data (76,788 Bytes)*
23'h12CFC	*CRC for Demura LUT (2 Bytes)*
23'h12CFE	*Dummy*

Demura Mode1
(RGB Mode, Plane 3, Compression)

Address	Data
23'h00000	*Header (4 Bytes)*
23'h00004	*Demura CFG Data (256 Bytes)*
23'h00104	*CRC for header & CFG (2 Bytes)*
23'h00106	*Dummy*
23'h00108	*Demura LUT Data (73,476 Bytes)*
23'h1200C	*CRC for Demura LUT (2 Bytes)*
23'h1200E	*Dummy*

Demura Mode2
(White Mode, Plane 3, No Compression)

- **Example**
  - Demura Mode 2

```
Offset(h)  00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

00000000   01 01 01 01 31 12 02 00 FF 03 FF 03 FF 00 00 00
00000010   00 FF FF A0 A0 00 C0 01 7A 02 80 00 00 00 80 80
      ⋮
00000100   00 00 00 00 93 C6 81 7F 81 7F 81 7F 81 7F 81 7F
00000110   81 7F 81 7F 81 7F 81 7F 81 7F 81 7F 81 7F 81 7F
      ⋮
0000F750   7F 81 7F 81 7F 81 7F 81 7F 81 00 00 00 00 00 00
0000F760   00 00 7F 81 7F 81 CD C2
```

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```
char* LX89507_Demura_main(double**** img, wchar_t* lut_output_path);
```

Arguments	I/O	Description
img	I	RGB Raw Camera Data. - Format : double **** [plane][color][x][y] - All Camera Data from both planes and colors is stored as an 4-dimensional array. - DMR_MODE is a register value < DMR_MODE 0 > [plane0][R] [plane0][G] [plane0][B] [plane1][R] [plane1][G] [plane1][B] < DMR _MODE 1 > [plane0][R] [plane0][G] [plane0][B] [plane1][R] [plane1][G] [plane1][B] [plane2][R] [plane2][G] [plane2][B] < DMR_MODE 2 > [plane0][W] [plane0][W] [plane0][W] [plane1][W] [plane1][W] [plane1][W] [plane2][W] [plane2][W] [plane2][W]
output_path		Max.4096 bytes. LUT data(.bin) is saved at output_path. [example] output location → ./_out/LUT.bin output_path → ./_out/LUT - The last part of a path should be ended with **the name of the output data**
return	O	Total CRC value

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Demura Library Interface Guide

Method of creating a block-sized image

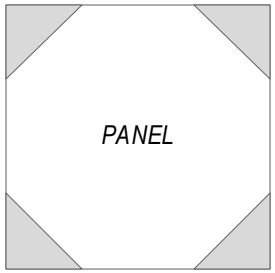
- *lxs_para.ini*

Parameter	Default	Min	Max	Description
IMG_BLOCK	1	0	1	Input image selection mode 0: Full resolution Image 1: 16x16 blocked Image

- *IMG_BLOCK == 0*
  - *Camera Image → Corner cut processing → Offset Calculation → Padding → Group with Block Size → Other Calculation*
  - *Camera Image : Full Resolution Image*

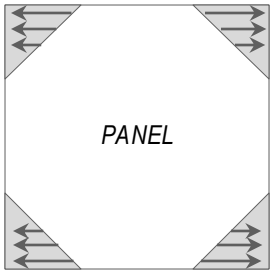
- *IMG_BLOCK == 1*
  - *Camera Image → Offset Calculation → Other Calculation*
  - *Camera Image : 16x16 Blocked Image (Corner cut processed)*

- *How to make Camera Image when IMG_BLOCK == 1*



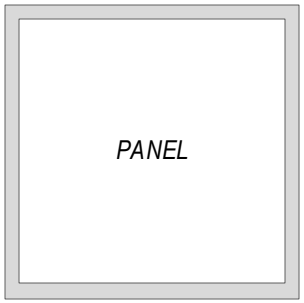
EX) 2488 x 2328

Original Camera Image



EX) 2488 x 2328

1. Corner cut processing
Replace all the values in the corner cuts of the panel with the values of the corners



EX) 2512 x 2352

2. Padding
To group the image data with a block size, padding is applied to adjust the resolution of the image to a multiple of the block size.

EX) 157 x147

3. Group with block size
Take the average or median of 1x1 data by block size x block size.
[16x16 Blocked Image size]
- *ceil(2488 / 16) + 1 = 157*
- *ceil(2328 / 16) + 1 = 147*

Demura Library Interface Guide

[Example] LX89507_Demura_main.cpp

```
char* check_demura;
int height = 2488;
int width = 2328;
std::string path = "./LX89507_Demura.dll";

HMODULE hDll = ::LoadLibraryA(path.c_str());

{
    DLL_MAIN Func1 = (DLL_MAIN)::GetProcAddress(hDll, "LX89507_Demura_main");

    wchar_t lut_output[4096] = L"./_out/LUT_out";
    wchar_t* lut_output_path = lut_output;

    check_demura = Func1(img, lut_output_path);
    if (check_demura)   printf("₩n₩n₩n========= LX89504 DEMURA SUCESSFULLY FINISHED !=========");

    ::FreeLibrary(hDll);
}
```

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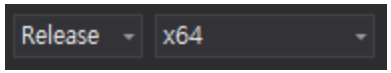
Demura Library Platform



x64 : for 64bit machine

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lxs_reg.ini (register)

Parameter	Default	Min	Max	Description
DMR_EN	0	0	1	Demura enable 0 : Demura Disable 1 : Demura Enable
DMR_MODE	2	0	2	Demura mode 0 : 16x16 (2 PLANE w/MDC, Compression Ratio 2:1) 1 : 16x16 (3 PLANE w/MDC, Compression Ratio 3:1) 2 : 16x16 (3 PLANE w/o MDC, white mode)
TEMP_GAIN_EN	0	0	1	Temperature gain enable 0 : Disable 1 : Enable
CONV_HIGH_EN	1	0	1	Converge enable of high area 0 : Converge Disable 1 : Converge Enable
CONV_LOW_EN	1	0	1	Converge enable of low area 0 : Converge Disable 1 : Converge Enable
SMOOTHING_EN_P0, P1, P2	1	0	1	Smoothing enable in Plane0, 1, 2
CONV_HIGH_STT	959	0	1023	Converge High Start CONV_LOW_STT < CONV_HIGH_STT < CONV_HIGH_END (HIGH_END – HIGH_STT) should be 2^n, where n is an integer
CONV_HIGH_END	1023	0	1023	Converge High End CONV_HIGH_STT < CONV_HIGH_END ≤ 1023 (HIGH_END – HIGH_STT) should be 2^n, where n is an integer
CONV_LOW_STT	64	0	1023	Converge Low Start CONV_LOW_END < CONV_LOW_STT < CONV_HIGH_STT (LOW_STT – LOW_END) should be 2^n, where n is an integer
CONV_LOW_END	0	0	1023	Converge Low End 0 ≤ CONV_LOW_END < CONV_LOW_STT (LOW_STT – LOW_END) should be 2^n, where n is an integer

0 ≤ CONV_LOW_END < CONV_LOW_STT< CONV_HIGH_STT < CONV_HIGH_END

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lxs_reg.ini (register)

Parameter	Default	Min	Max	Description
MIN_GRAY_DIFF	255	0	255	Extrapolation point for 0 gray
MAX_GRAY_DIFF	255	0	255	Extrapolation point for 255 gray
FBIT_REGION1	160	0	2560	Region1 block vertical position for Fraction bit
FBIT_REGION2	160	0	2560	Region2 block vertical position for Fraction bit
PLANE00, PLANE01, PLANE02	128, 256, 512	0	1023	10bit Gray value (RGB) of each plane
PLANE00_FBIT_R1, R2 PLANE01_FBIT_R1, R2 PLANE02_FBIT_R1, R2	0	0	15	Plane's fraction bit 0 : 8s0, 1 : 7s1, 2 : 6s2, 3: 5s3, 4 : 4s4, 15 : 9s-1
PLANE00_TEMP_R_GAIN PLANE00_TEMP_G_GAIN PLANE00_TEMP_B_GAIN	128	0	255	Temperature R/G/B Gain for Plane 0 (x1 : 128, Range : x0.0 ~ x1.9921875)
PLANE01_TEMP_R_GAIN PLANE01_TEMP_G_GAIN PLANE01_TEMP_B_GAIN	128	0	255	Temperature R/G/B Gain for Plane 1 (x1 : 128, Range : x0.0 ~ x1.9921875)
PLANE02_TEMP_R_GAIN PLANE02_TEMP_G_GAIN PLANE02_TEMP_B_GAIN	128	0	255	Temperature R/G/B Gain for Plane 2 (x1 : 128, Range : x0.0 ~ x1.9921875)

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lxs_reg.ini (register)

Parameter	Default	Min	Max	Description
APPR_BASE_PLANE	1	0	2	Offset Approximation Base Plane number (Base offset selector)
APPR_TARGET_PLANE	2	0	2	Offset Approximation Target Plane number (Target Offset position)
OFST_APRM_GAIN_R OFST_APRM_GAIN_G OFST_APRM_GAIN_B	64	0	255	Offset Approximation Gain R/G/B (x1 : 64, Range : x0.0 ~ x3.984375)
FRD_OPT	2	0	2	FRD option in smoothing algorithm 0 : Even line 1 : Odd line 2 : Average
FRD_OFST_P0_R FRD_OFST_P0_G FRD_OFST_P0_B	0	0	255	FRD Offset Plane 0 Red, Green, Blue
FRD_OFST_P1_R FRD_OFST_P1_G FRD_OFST_P1_B	0	0	255	FRD Offset Plane 1 Red, Green, Blue
FRD_OFST_P2_R FRD_OFST_P2_G FRD_OFST_P2_B	0	0	255	FRD Offset Plane 2 Red, Green, Blue
FRD_OFST_FBIT_P0_R FRD_OFST_FBIT_P0_G FRD_OFST_FBIT_P0_B	0	0	255	Fraction bit for FRD Offset Plane 0 Red, Green, Blue 0 : 6s2, 1 : 5s3, 2 : 4s4, 3: 3s5
FRD_OFST_FBIT_P1_R FRD_OFST_FBIT_P1_G FRD_OFST_FBIT_P1_B	0	0	255	Fraction bit for FRD Offset Plane 1 Red, Green, Blue 0 : 6s2, 1 : 5s3, 2 : 4s4, 3: 3s5
FRD_OFST_FBIT_P2_R FRD_OFST_FBIT_P2_G FRD_OFST_FBIT_P2_B	0	0	255	Fraction bit for FRD Offset Plane 2 Red, Green, Blue 0 : 6s2, 1 : 5s3, 2 : 4s4, 3: 3s5

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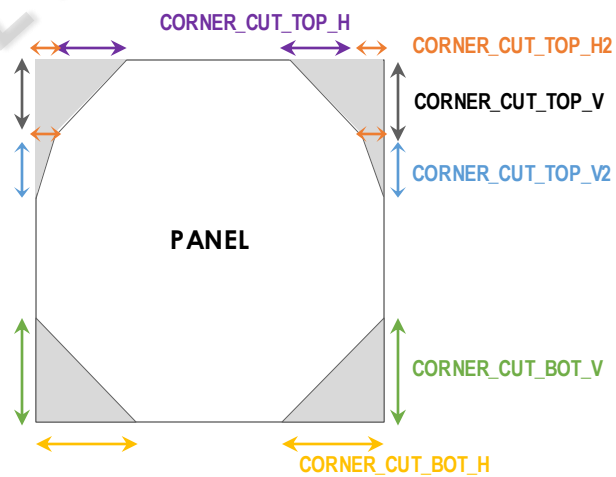
lxs_para.ini (parameter)

Parameters are only used in a DLL (SW encoder) to create a LUT.

Parameter	Default	Min	Max	Description
HEIGHT	2488	0	2488	Input Height (Panel Vertical Width)
WIDTH	2328	0	2328	Input Width (Panel Horizontal Width)
BLOCK_MEDIAN	1	0	1	Method for converting 1x1 data to block size x block size 0: Average of the data 1: Median of the data
TARGET_RADIUS	111	1	-	To determine the target value for Demura, a circular area is selected at the center of the panel. The average value of the pixels within the circle is used as the target value, and the radius of the circle is also determined as TARGET_RADIUS.
CORNER_CUT_TOP_H	420	0	-	The distance between a vertex of the diagram that represents the panel and the corner of the rectangle that encloses the panel. Please refer to the image below. 
CORNER_CUT_TOP_V	420	0	-	
CORNER_CUT_TOP_H2	145	0		
CORNER_CUT_TOP_V2	342	0	-	
CORNER_CUT_BOT_H	487	0	-	
CORNER_CUT_BOT_V	489	0	-	

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lxs_para.ini (parameter)

Parameters are only used in a DLL (SW encoder) to create a LUT.

Parameter	Default	Min	Max	Description
BLOCK_SIZE	16	1	-	Block size (LX89507 Cassandra Demura IP only supports BLOCK_SIZE==16)
IMG_BLOCK	1	0	1	Input image selection mode 0: Full resolution Image 1: 16x16 blocked Image
OUTPUT_TXT	1	0	1	Whether the result is saved as a txt file. 0 : The LUT and registers are saved in a '.bin' file. 1 : The LUT and registers are saved in a '.bin' file, and also in separate '.txt' files.
FBIT_AUTO	1	0	1	Fraction bit setting method selection mode 0: Manually fix the FBIT using the value in 'LXS_reg.ini'. 1: Automatically set up the FBIT in the encoder.
TOTAL_CRC	1	0	1	TOTAL_CRC file generation ON/OFF option 0: Total_crc.bin is not generated 1: Total_crc.bin is generated

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